

2(a)	• resistance of a metal • volume of a gas at constant pressure • e.m.f. of a thermocouple <i>Any two points, 1 mark each</i>	B2
2(b)(i)	$Q = mc\Delta T$	C1
	evidence of realisation that Q lost by water = Q gained by mercury	C1
	$18.7 \times 4.18 \times (37.4 - T) = 6.94 \times 0.140 \times (T - 23.0)$	C1
	$T = 37.2\text{ }^{\circ}\text{C}$	A1
2(b)(ii)	use a liquid with a lower (specific) heat capacity (than mercury) or use a smaller mass of mercury	B1
2(c)(i)	depends on properties of a real substance	B1
	$0\text{ }^{\circ}\text{C}$ is not absolute zero	B1
2(c)(ii)	ideal gas	B1